

Periods, quasi-periods, and where the determinant comes from

An elliptic-curve / modular-form Rosetta stone

Working note — conceptual, not manuscript text

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Goal: explain why $\omega^+ \eta^- - \omega^- \eta^+$ is a power of $2\pi i$ (up to rationals), and why Brown’s words — “period,” “quasi-period,” “first kind,” “second kind” — are the sensible ones. The whole story is one structural fact (a 2×2 period matrix between a de Rham and a Betti basis, with a symplectic pairing) seen twice: once for an elliptic curve, where it is the classical Legendre relation, and once for Δ , where it is Brown’s determinant. The determinant is an intersection number, and that is why your construction should be able to derive it rather than merely match it numerically.

1 The elliptic curve, concretely (the case to anchor on)

Let E be an elliptic curve over $\mathbb{Q}$. Its first algebraic de Rham cohomology $H_{\text{dR}}^1(E)$ is 2-dimensional, and it has a distinguished basis of *differentials*:

$$\omega = \frac{dx}{2y+1} \quad (\text{first kind}), \quad \eta = \frac{x dx}{2y+1} \quad (\text{second kind}).$$

“First kind” means *holomorphic* (no poles anywhere, including at infinity); it is the everywhere-regular differential. “Second kind” means it has poles but *no residues* — so it still defines a cohomology class (a residue would make the period path-dependent / ill-defined). η has a double pole at the origin of E and no residue. The pair (ω, η) is a basis of H_{dR}^1 , and the second basis vector is genuinely *not* holomorphic — that is the entire point of needing two of them.

On the Betti side, $H_1(E(\mathbb{C}); \mathbb{Z}) \cong \mathbb{Z}^2$ has a basis of *cycles* (γ_1, γ_2) (the two loops of the torus). Pairing differentials against cycles — integrate a form over a loop — gives the **period matrix**

$$P = \begin{pmatrix} \int_{\gamma_1} \omega & \int_{\gamma_1} \eta \\ \int_{\gamma_2} \omega & \int_{\gamma_2} \eta \end{pmatrix} = \begin{pmatrix} \omega_1 & \eta_1 \\ \omega_2 & \eta_2 \end{pmatrix}.$$

The left column (ω_1, ω_2) are the **periods** — the integrals of the holomorphic differential, the lattice generators. The right column (η_1, η_2) are the **quasi-periods** — the integrals of the second-kind differential. (Classically, with $\omega = dz$ and $\eta = \wp(z) dz$ in Weierstrass coordinates, the η_i are the values of the Weierstrass ζ -function increments; “quasi-period” is the period of the second-kind partner of the period.)

In one line. *Periods integrate the first-kind (holomorphic) differential; quasi-periods integrate its second-kind (meromorphic, residue-free) partner. Both are columns of one 2×2 period matrix.*

2 Legendre’s relation *is* the determinant

The determinant of P is not a generic transcendental — it is forced:

$$\boxed{\det P = \omega_1 \eta_2 - \omega_2 \eta_1 = 2\pi i} \quad (\text{Legendre’s period relation}).$$

(With the normalization $(\omega, \eta) = (dx/(2y+1), x dx/(2y+1))$ this is exactly $2\pi i$; in Fonseca’s worked $\mathbb{Q}$ -curve it is verified numerically to $\det P = 2\pi i$ on the nose.)

Why is the determinant special, when the individual entries are honest transcendentals (the ω_i alone are not algebraic)? Because $H_{\text{dR}}^1(E)$ carries a **symplectic pairing** — the cup product / the algebraic intersection pairing on H^1 — and $\langle \omega, \eta \rangle$ is a *rational* (here, essentially integer) number, since it is computed algebraically from the differentials. The Betti side carries the dual pairing, the topological intersection number $\gamma_1 \cdot \gamma_2 = 1$ of the two torus loops. The comparison isomorphism $H_{\text{dR}}^1 \otimes \mathbb{C} \cong H_{\text{B}}^1 \otimes \mathbb{C}$ — “integrate the form over the cycle,” i.e. the matrix P — must *intertwine* the two pairings, and the factor by which a comparison isomorphism scales a symplectic form is its determinant. Concretely:

$$\underbrace{\langle \omega, \eta \rangle_{\text{dR}}}_{\in \mathbb{Q}} = \frac{1}{2\pi i} \det P \cdot \underbrace{(\gamma_1 \cdot \gamma_2)}_{=1}.$$

So $\det P$ is pinned to $2\pi i$ times a rational *because* it is the ratio of two rational symplectic volumes (de Rham and Betti), with the $2\pi i$ the normalization of the comparison on H^1 of a curve. The determinant is an **intersection number in disguise**; that is the whole content of Legendre.

In one line. $\det P = 2\pi i$ holds because P matches a rational de Rham symplectic pairing to a rational Betti one; the determinant is the scaling factor, and $2\pi i$ is the normalization of the comparison on a curve. Legendre = “the periods respect the intersection form.”

3 The modular form, same skeleton

Now replace the curve by the *family* of all elliptic curves — the modular curve / moduli stack $\mathcal{M}_{1,1}$ — and take symmetric powers of the rank-2 bundle. Brown’s Theorem (1710.07912 §2; QP §1) is the family version of “ $H_{\text{dR}}^1(E)$ is 2-dimensional with a first- and second-kind basis”:

$$M_{n+2}^!(\mathbb{Q})/D^{n+1}M_{-n}^!(\mathbb{Q}) \xrightarrow{\sim} H_{\text{dR}}^1(\mathcal{M}_{1,1}; \text{Sym}^n), \quad D = q \frac{d}{dq}.$$

Read this slowly, because it is the crux. The left side is weakly holomorphic modular forms of weight $n+2$, *modulo* $(n+1)$ -fold q -derivatives of weight- $(-n)$ forms (Bol’s operator D^{n+1} raises weight $-n$ to $n+2$). The quotient is what Brown calls **modular forms of the second kind**, and the theorem says it *is* the de Rham cohomology of the local system. So:

Elliptic curve	Modular ($k = 12, n = 10$)
$H_{\text{dR}}^1(E)$, rank 2	$H_{\text{dR}}^1(\mathcal{M}_{1,1}; \text{Sym}^{10})$, rank 2 per Hecke eigenspace
$\omega = dx/(2y+1)$, first kind (holomorphic)	$\Delta \in S_{12}$, the holomorphic cusp form
$\eta = x dx/(2y+1)$, second kind (pole, no residue)	$\hat{\Delta} \in S_{12}^!$, weakly holomorphic — the second-kind partner
de Rham symplectic pairing $\langle \omega, \eta \rangle \in \mathbb{Q}$	the Poincaré / cup pairing on $H_{\text{dR}}^1(\text{Sym}^{10})$, $\mathbb{Q}$ -valued
Betti cycles $\gamma_1, \gamma_2, \gamma_1 \cdot \gamma_2 = 1$	the cuspidal cohomology $H_{\text{cusp}}^1(\Gamma; V_{10})$, rank 2
period matrix $P = \left(\int_{\gamma_i} \omega \mid \int_{\gamma_i} \eta \right)$	$P_f = \begin{pmatrix} \eta^+ & \omega^+ \\ i\eta^- & i\omega^- \end{pmatrix}$ (Brown)
$\det P = 2\pi i$ (Legendre)	$\det P_f \doteq (2\pi i)^{k-1}$ (up to $\mathbb{Q}^\times$)

The dictionary is not analogy-by-vibes; it is literally the same theorem applied to Sym^n of the curve's H^1 instead of H^1 itself. The left ($\omega^\pm$) column integrates the *holomorphic* representative Δ — these are Eichler–Shimura's classical periods, your $\int_0^{i\infty} \Delta(X - \tau Y)^{10} d\tau$. The right ($\eta^\pm$) column integrates the *second-kind* representative $\hat{\Delta}$ — the quasi-periods. “Quasi-period” is the right word for exactly the reason it is on the curve: it is the period of the second-kind partner of the period.

In one line. $\Delta : \hat{\Delta} = \omega : \eta = \text{first kind} : \text{second kind}$. The quasi-periods of $\hat{\Delta}$ are the modular analog of the quasi-periods $\eta_i = \int_{\gamma_i} x dx/y$ of an elliptic curve. Same matrix, same determinant story, one symmetric power up.

4 So where does the modular determinant come from?

Exactly where Legendre's does, with one distinction worth stating up front because it is easy to get wrong: the relevant pairing is the *determinant of the period matrix* P_f — the comparison isomorphism between de Rham and Betti realizations — and this is **not** the same as the cup product on cohomology. (The cup product is a different invariant; see the box below.) The determinant of P_f is the scaling factor by which the comparison isomorphism relates the rational de Rham structure to the rational Betti structure, and that scaling is forced to be a rational multiple of the normalizing $(2\pi i)^{k-1}$. The power is $k-1 = n+1$ because the realizations live on Sym^n of H^1 , not H^1 .

Numerically (next section), the rational is exactly $n! = 10!$:

$$\omega^+ \eta^- - \omega^- \eta^+ = n! (2\pi)^{n+1} = 10! (2\pi)^{11} \quad (k = 12).$$

In Brown's normalization the $\omega^\pm$ are defined only up to $K^\times$ (the Hecke field) and the $\eta^\pm$ up to $K^\times$ and up to adding K -multiples of $\omega^\pm$ — which is exactly the statement that the second-kind class is well-defined only modulo the first-kind one (you can always add a holomorphic form to a second-kind form). That “modulo $\omega^\pm$ ” freedom is the *same* ambiguity as your $\mathbb{C}(X^n - Y^n) / \mathbb{C}Y^n$ slack: it is the de Rham analog of “a second-kind differential is defined up to first-kind differentials and exact forms.”

Determinant vs. cup product — do not conflate them (a correction). An earlier version of this note claimed the determinant $\omega^+\eta^- - \omega^-\eta^+$ is the cup pairing $\langle [C^\Delta], [C^{\hat{\Delta}}] \rangle$ on H^1 . That is wrong, and a direct computation shows why. The invariant symplectic pairing $\langle \cdot, \cdot \rangle$ on V_n *diagonalizes by parity*: it couples the even part with the even part and the odd with the odd (numerically, $\langle B^+, B^- \rangle = 0$ for the Zagier bracket polynomials). So the cup product of two period polynomials is a *parity-diagonal* combination, $\omega^+\eta^+ \langle B^+, B^+ \rangle - \omega^-\eta^- \langle B^-, B^- \rangle$, which couples $+$ with $+$. The determinant couples $+$ with $-$. They are different bilinear forms in the same four numbers, and no normalization turns one into the other.

What each one computes: the *cup product* (Haberland’s formula) is proportional to the *Petersson norm* — for $f = g = \Delta$ it returns $i(\Delta, \Delta)$, matching Zagier’s $\omega^+\omega^-/i(\Delta, \Delta) = 2^{10}$ up to the V_n -pairing normalization. The *determinant* of P_f is the comparison-isomorphism volume ratio — the Legendre/quasi-period relation, $= 10!(2\pi)^{11}$. Both are real, finite, computable; they are simply different invariants, one diagonal and one cross in the parity grading.

Two cross-checks on the determinant reading, both from your project knowledge:

(i) *Zagier 1991*, $k = 12$. Zagier records $\omega_\Delta^+\omega_\Delta^-/i(\Delta, \Delta) = 2^{10} \in \mathbb{Q}$, where (Δ, Δ) is the Petersson norm. This is the *diagonal* (cup-product / Petersson) statement — the partner of, but distinct from, the off-diagonal determinant. It confirms the rational-times-normalization pattern on the diagonal.

(ii) *Brown 1710.07912 §1.3*. The “single-valued period” $s(f)$ is built as $P^{-1}\bar{P}$ (Fonseca eq. 4), and Brown’s eq. (1.6) writes $s(f)$ with denominator $\eta^-\omega^+ - \eta^+\omega^- = \det P_f$ (up to sign/i), explicitly noting $\det(P_f)$ is proportional to a power of $2\pi i$ and that $\omega^+\omega^-$ is related to the Petersson norm. The determinant appears there precisely as the normalizing denominator of a comparison-isomorphism construction — i.e. as the symplectic scaling factor.

5 What this does and doesn’t pin down

If the determinant were a free parameter you could tune, matching it would mean something. It isn’t: $\det P_f = (\text{rational}) \cdot (2\pi i)^{k-1}$ is a structural identity (the de Rham/Betti comparison volume ratio), so *any* four numbers that are genuine periods/quasi-periods satisfy it. This cuts both ways. If your $\omega^\pm, \eta^\pm$ match Brown’s tabulated values — which they do, to 20 digits — then the determinant relation holds automatically, and “verifying” it is checking Brown against Brown through your arithmetic. It is empty as a test of either Brown or your construction.

I previously proposed computing the cup pairing $\langle [C^\Delta], [C^{\hat{\Delta}}] \rangle$ from cocycle data as an *independent* route to the determinant. That proposal was based on a conflation, now corrected (see the box in §4): the cup pairing is parity-diagonal and computes the *Petersson norm*, not the *cross-parity* determinant. So computing the cup product does not give an independent handle on $\omega^+\eta^- - \omega^-\eta^+$; it gives $i(\Delta, \Delta)$ instead. There is no quick cocycle-pairing check of the determinant, because the determinant simply is not that pairing.

What *would* carry independent information is more pedestrian and you are already most of the way there: your nine-way agreement across distinct basepoints τ_0 already stresses the extraction pipeline hard, and the fact that the *same* finite-contour formula yields $\omega^\pm(\Delta)$ and $\eta^\pm(\hat{\Delta})$ by only changing the integrand is the real structural content — that is what places periods and quasi-periods on one footing, and it is verified by the construction itself, not by any external pairing.

The constant, pinned numerically. From Brown’s four $k = 12$ values one finds, to 16 digits,

$$\omega^+\eta^- - \omega^-\eta^+ = 10!(2\pi)^{11} = n!(2\pi)^{n+1} \quad (n = 10),$$

i.e. the “rational $\times$ normalization” constant is exactly the factorial $n! = 3628800$ (with $(2\pi)^{n+1}$; equivalently $\pm n! (2\pi i)^{n+1}$, the sign/ i depending on the convention carried for the imaginary ω^-, η^-). This used only the four numerical values and $(2\pi)^{11}$ — no V_n -pairing normalization entered it, so the $n!$ is solid. It is not a test of Brown (these are his numbers, and the relation is a theorem they satisfy by construction); it merely records the specific rational, in case you state the Legendre relation in print.

The note in four sentences. An elliptic curve has a rank-2 de Rham cohomology with a first-kind (holomorphic) differential dx/y and a second-kind (residue-free meromorphic) partner $x dx/y$; integrating them over the two torus cycles gives the period matrix, whose determinant is Legendre’s $2\pi i$ because the matrix intertwines the rational de Rham and Betti intersection pairings. Brown’s theorem says weakly holomorphic modular forms modulo D^{n+1} (lower-weight) *are* the de Rham cohomology of Sym^n of that family, with Δ the first-kind class and $\hat{\Delta}$ its second-kind partner, so $\omega^\pm$ (periods of Δ) and $\eta^\pm$ (quasi-periods of $\hat{\Delta}$) are the two columns of the same kind of period matrix, one symmetric power up. The determinant $\omega^+ \eta^- - \omega^- \eta^+ = n! (2\pi)^{n+1}$ (numerically $10! (2\pi)^{11}$ at $k = 12$) is the modular Legendre relation: the de Rham/Betti comparison volume, forced to be rational $\times (2\pi i)^{k-1}$. This is distinct from the cup product on cohomology, which is parity-diagonal and computes the Petersson norm instead — so there is no shortcut “cup-pairing” check of the determinant, and matching it against Brown’s values is empty (the relation holds for any genuine periods by construction); the real structural content of your construction is that one finite-contour formula yields both $\omega^\pm(\Delta)$ and $\eta^\pm(\hat{\Delta})$.